

Chapter 4: levels of prevention

There are three levels of prevention—primary, secondary and tertiary

Primary prevention

This is the measure undertaken in the period of **pre pathogenesis (i.e. before the development of the disease)** which removes the possibility of occurrence of the disease. Especially, it is all the more important in those diseases, for which no treatment is available, e.g. AIDS, cancer, rabies, caries tooth, etc.

Primary prevention can be adopted by two modes of intervention—namely Health promotion and Specific protection.

Health Promotion

This consists of ‘general measures’, which will strengthen the individual/host and prevents the occurrence of the disease by interrupting the interaction among the three factors of epidemiological triad. Examples;

- Health education
- Adequate nutrition
- Improvement in the environmental sanitation (such as control of insects, provision of protected water supply, sanitary disposal of sewage, etc

Specific Protection

This consists of ‘specific measures’, which prevent specific diseases. Examples;

- Immunization against vaccine preventable diseases

- Silver nitrate or penicillin eye drops against ophthalmia neonatorum
- Condom against AIDS
- Helmet against head-injury

Secondary prevention

- This is the measure undertaken in the early stage, **after the onset of the disease (in the early pathogenesis phase).**
- The intervention is by **‘Early diagnosis and treatment ex.,**
 1. Periodic recording of a weight of the child to detect malnutrition
 2. Screening of blood donors to detect HIV/AIDS, hepatitis B, syphilis, malaria
 3. Recording of blood pressure to detect hypertension.

Different modalities of treatment are chemotherapy, radiotherapy, immunotherapy, hormonal therapy, psychotherapy, physiotherapy, oral rehydration therapy and surgery. Conditions where the treatment is for a long period as in tuberculosis or leprosy, the physician should ensure ‘Case holding’, i.e. to see that the patient takes the treatment correctly and completely.

Tertiary prevention

- This is the measure undertaken, **when the disease process is sufficiently advanced, (i.e. in the late pathogenesis phase).**
- It includes two modes of intervention; Disability limitation__ and Rehabilitation.

1. Disability Limitation:- This means limiting the development of further disability in the individual by giving intensive or aggressive treatment

2. Rehabilitation:-

- Combined and use of physical, social, vocational and psychological measures for retraining the individual to the highest possible level of functional ability', so that the individual becomes useful to himself, to the family and to the community at large.
- Physical rehabilitation→ ex., artificial limb.
- Vocational rehabilitation → ex., employment
- Social rehabilitation → Restoration of relationship in the society
- Psychological rehabilitation → Restoration of personal confidence

The sequence of events in a disease process are:

➤ ***Disease → Impairment → Disability → Handicap***

Disease or event (accident)→ Impairment (loss of organ structure or function e.g., loss of foot) → Disability (Inability to perform normal daily activities e.g., inability to walk) → Handicap (loss of social role e.g., Loss of job or unemployment)

References

1. Suryakantha AM., *Community Medicine with recent advances. Vol 3. Chapter 4. Jaypee Brothers Medical Publishers, New-Delhi, India, 2014.*